

HAT-P-32b

R-1259

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_211221 · created 2026-07-31T12:33:23+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459569.723 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.013
Transit depth (Rp/Rs)^2	2.38 +/- 0.4 [%]
Orbital Inclination (inc)	85.56 +/- 1.68
Airmass coefficient 1 (a1)	1.031 +/- 0.016
Airmass coefficient 2 (a2)	-0.024 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	#3 - [385, 244]
Optimal Aperture	3.23
Optimal Annulus	14.28
Transit Duration (day)	0.1239 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.013 vs 0.1489 (deviation 0.39σ)
Duration fit vs published	0.1239 d vs 0.1304 d (deviation 1.18σ)
Mid-time O–C	9.3 min
Depth SNR	11.8
Residual scatter	1.53 %

Light curve

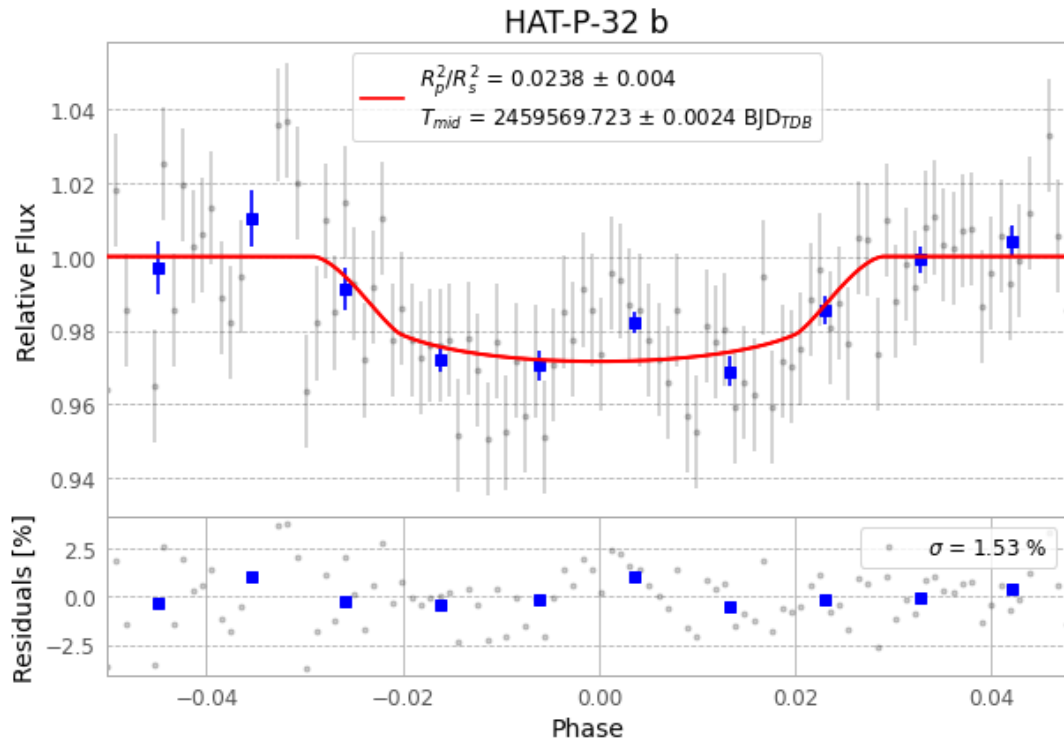


Figure 1 — FinalLightCurve_HAT-P-32 b_2021-12-20.png (SHA-256 d0aa2dfab3b73783...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 253], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ef2fe690e1c4bf75...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

102 FITS frames (67 MB), HATP-32211221023912.FITS → HATP-32211221074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-211221.log (fb6ed6187f00...), FinalLightCurve_HAT-P-32 b_2021-12-20.png (d0aa2dfab3b7...), FinalLightCurve_HAT-P-32 b_2021-12-20.csv (e404e66a3c08...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:33Z.